
Keyed PRNG selects carrier blocks and chips
Pixel array: per-region gain ramp ($\pm A$ chip)
ADC + ISP readout, encode $E(I)$
Sign record r with on-die sk
Merkle batch $h = \mathscr{H}(r \,\|\, \psi)$
Public ledger (seal root)
carrier key k (OEM)
payload $\mathbf{m} \rightarrow$ RS code (ID, time, location)
sensor key sk (certified in lot Λ)
soft binding: payload carried in the pixels
hard binding: signed record sealed on chain